



ENERGETIKA
ENERGETICS

Published from 2004
Ministry of Press and Information
of Azerbaijan Republic,
Registration number 3337, 07.03.2011

ISSN 1816-2126
Number 01, 2025
Section: English

ECOENERGETICS

HONORARY EDITOR IN CHIEF: Fagan G. Aliyev
SENIOR EDITOR: Rashad G. Abaszade
MANAGING EDITOR: Elmira A. Khanmamadova

INTERNATIONAL REVIEW BOARD

Arif Pashayev, Azerbaijan	Rafiq Aliyev, Azerbaijan	Turhan Vaziroglu, USA
Vagif Abbasov, Azerbaijan	Fuad Hajizadeh, Azerbaijan	Shiro Takada, Japan
Vagif Farzaliyev, Azerbaijan	İsmayil Aliyev, Azerbaijan	Luca Di Palma, İtalia
Natig Cavadov, Azerbaijan	Nazim İmanov, Azerbaijan	Yuriy Tabunshikov, Russian
Khadiyya Khalilova, Azerbaijan	Ali Guliyev, Azerbaijan	Mithat Kaya, Turkey
Farhad Aliyev, Azerbaijan	Leyla Mammadova, Azerbaijan	Elvin Aliyev, UK
Sakin Cabarov, Azerbaijan	Nazim Shamilov, Azerbaijan	Emre Gür, Turkey
Adil Azizov, Azerbaijan	Salahaddin Yusifov, Azerbaijan	Volodymyr Kotsyubynsky, Ukraine
Azer Mammadov, Azerbaijan	Mazahir İsayev, Azerbaijan	Matlab Mirzayev, Russian
Nurmammad Mammadov, Azerbaijan	Yusif Aliyev, Azerbaijan	Olga Kapush, Ukraine
Akif Alizadeh, Azerbaijan	Adil Abdullayev, Azerbaijan	Aitbek Aimukhanov, Kazakhstan
Rahim Alakbarov, Azerbaijan	Alakper Hasanov, Azerbaijan	Maksym Stetsenko, China
Gurban Eyyubov, Azerbaijan	Sevinj Malikova, Azerbaijan	Aida Bakirova, Kyrgyzstan
Samad Yusifov, Azerbaijan	Tarlan Huseynov, Azerbaijan	Bakhtyar Soltabayev, Kazakhstan
Ruslan Nuriyev, Azerbaijan	Tamella Naibova, Azerbaijan	Dzmitry Yakimchuk, Belarusiya
Agali Guliyev, Azerbaijan	Oktay Salamov, Azerbaijan	Saifullah Khalid, İndia
Tural Nagiyev, Azerbaijan	Asif Pashayev, Azerbaijan	Anitha Velu, İndia
Latif Aliyev, Azerbaijan	İlman Hasanov, Azerbaijan	Koushik Guha, India
		Nitesh Dutt, India
		Neeraj Kumar Bhoi, India
		Akbar Haghi, Portugal
		Muhammad Sultan, Pakistan

TECHNICAL EDITORIAL BOARD

SENIOR SECRETARY: İmran Y. Bayramov, Afig M. Nabiyev, Karim G. Karimov, Turan A. Nahmatova,
Nigar V. Abbasova, Rashid Y. Safarov, Nuranə A. Zohrabbayli, Seynura A. Hasanova, Kamila A. Cafarli.

PUBLISHING OFFICE

5, M.Rahim, AZ-1073, Baku Azerbaijan

Tel.: 99412 538-23-70, 99412 538-40-25 Fax: 99412 538-51-22

E-mail: info@ieeacademy.org ekoenergetics@innovationresearch.az

Internet: <http://ieeacademy.org> www.innovationresearch.az

CONTENTS

1	Advances and Trends in Micro- and Nano-Fabrication Rashad Abaszade, Elmira Khanmamadova	1-23
2	Application of drones to hybrid energy systems: operating cost reduction and advantages N.T.Vahabli	24-28
3	Economic and environmental studies in the sustainable development of renewable energy in Karabakh M.Huseynova	29-32
4	Comparative analysis of oil-contaminated soil remediation methods A.Azizov, S.Huseynli	33-35
5	Development of an Automated Control System for Crude Oil Heater and Its U.Rzayev, İ.Bayramov	36-38
6	Electromagnetic waves and their effects on the human organism T. Huseynov, B. Davudov	39-42
7	Investigating the impact of climate change on the caspian sea level and ecological conditions A.Khalilova, N.Farzaliyeva	43-48
8	Laser strengthening of CP65+5%MF abrasive material for gears of oil pumps A.Guliyev, Y.Jahangirova	49-56
9	Plastic Alternatives: Bioplastics and Nanotechnology L.H. Memmedova, F.F. Ismayilova	57-59
10	Rehabilitation, modernization and expansion of electrical networks in liberated territories K.Gulmammadov	60-68
11	Renewable energy sources in the territory of east zangezur and Karabakh S.Gasimova, G.Mammadova	69-71
12	Sustainable urbanization and green infrastructure for reducing carbon emissions S.Gasimova, N.Əhədzadə	72-75
13	Solutions for optimizing heat dissipation on the printed circuit board of the drive circuit of a DC motor Jamil Gunashov, Rashad Abaszade	76-82
14	Prompt-based forecasting of monthly solar irradiation in the fuzuli district using ai models Mobil Maharramov	83-86
15	Industrial waste processing and utilization: a study of Sumgayit technopark C.A.Gurbanov	87-90
16	Artificial Intelligence and Geology F.M.Hajizadeh, N.Y.Ahadi	91-95
17	Energy Sector Vulnerability Fidan Abdullazade, Farhad Aliyev	96-99
18	Water Degradation Challenges in Azerbaijan's Karabakh Region S. R. Yusifova	100-103
19	Waste collection and utilization at sumgait technologies park C.A.Gurbanov	104-106

Rehabilitation, modernization and expansion of electrical networks in liberated territories

K.Gulmammadov

Azerbaijan State Oil and Industry University, Azerbaijan

kenan200011@gmail.com

Abstract: Electrification is the pillar of the economic security of the society and the way of development of all areas of the national economy. Complete and high-quality electrification is carried out on the basis of modern technology and scientific achievements. Electricity production in the country is constantly increasing, improving and covering all areas of the national economy. Electricity production is developed in accordance with the energy program of the country. Electricity is produced through power plants and alternative energy sources. Centralized production of electricity ensures the reliability and economic feasibility of technological processes and works.

Keywords: electricity, network, energy, statistics, station

Corresponding authors, email: kenan200011@gmail.com

1.Introduction

The ratio of the energy produced by all generators to the energy received as a result of fuel combustion (in thermal power plants) or to the potential energy of water passing through the turbine (in thermal power plants) is called the efficiency of the station [11]. One of the indicators of the efficiency of a power plant is the cost of each kilowatt-hour of electricity produced. The average cost of energy per kilowatt-hour in the buses of a power plant is determined by the ratio of the annual costs of the power plant to the electricity produced. In Kalbajar, Lachin, Gubadli, Zangalan, Jabrayil and Fuzuli regions, there are 8 potential areas with a total area of 14 thousand 427 hectares and high solar energy potential. Although these areas can host solar power plants with a total capacity of 7,214 MW, the total potential here is estimated at more than 4,000 MW, considering that part of the southern areas can be used for agricultural purposes. Carbon-based nanomaterials are indispensable materials for energy transfer, storage, and production. Research has been conducted on their preparation, analysis, and applications [12-38].

2.Experimental detail

Regions with high renewable energy potential in the liberated territories should be considered separately, measurement and control stations should be installed in these regions in order to accurately determine the potential of wind and solar energy, proposals should be prepared for conducting work in the direction of evaluating technical capabilities and economic efficiency.

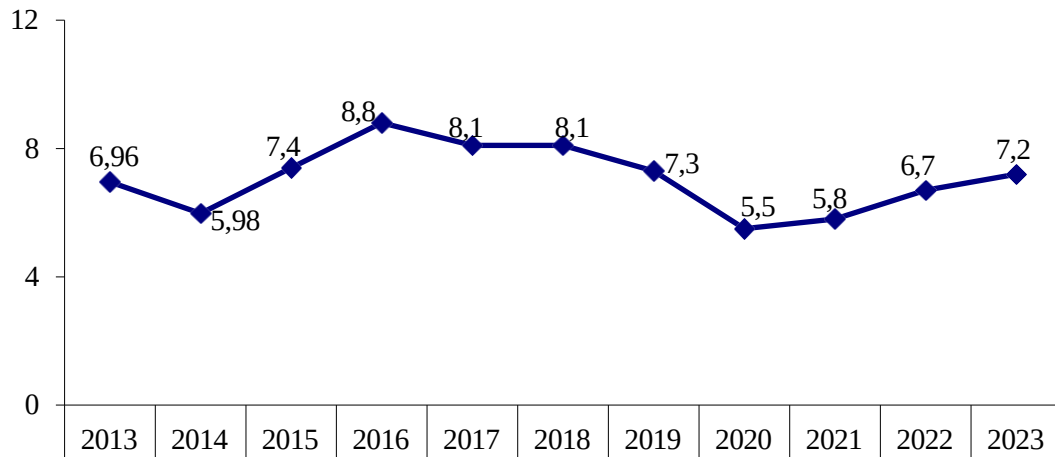


Fig 1. Percentage share of electricity from renewable energy sources in total electricity production in de-occupied territories

A power station usually has several generators. The power of all generators in operation and maintenance is called the installed power of the power plant. Currently, the total power of generators producing electricity is called operating power [4]. The power of generators that are not working for certain reasons, but can be started at any time, is called reserve power. When information about the amount of energy produced during a certain period of time is known, the average load of the plant is calculated using the following formula:

$$P_{or} = \frac{A}{T}$$

Here, P_{or} is the average load, A is the amount of energy produced, and t is the time.

Knowing the maximum value of the station load P_m for this period, we can determine the station load coefficient K_y . This factor is the ratio of the station load P_{or} to the maximum load:

$$K_y = \frac{P_{or}}{P_m}$$

Capacity of power plants, MW

Years	The capacity of power plants by the end of the year	including:					
		Fuel-fired ES and IES - total	SES	Wind ESE	The sun is ES	Solid household waste plant	Biogas ES
1913	39,8	39,8	-	-	-	-	-
1920	56,4	56,4	-	-	-	-	-
1930	113,4	113,4	-	-	-	-	-
1940	254,4	254,4	-	-	-	-	-
1950	401,6	401,6	-	-	-	-	-
1960	1.260,9	885,9	375,0	-	-	-	-
1970	2.622,5	2.247,0	375,5	-	-	-	-
1980	2.882,0	2.415,0	467,0	-	-	-	-

1990	5.050,7	4.263,0	787,7	-	-	-	-
1995	5.043,7	4.256,0	787,7	-	-	-	-
2000	4.911,9	3.990,0	921,9	-	-	-	-
2001	5.161,0	4.228,0	933,0	-	-	-	-
2002	5.283,0	4.349,0	934,0	-	-	-	-
2003	5.673,1	4.703,0	970,1	-	-	-	-
2004	5.665,1	4.695,0	970,1	-	-	-	-
2005	5.157,1	4.187,0	970,1	-	-	-	-
2006	5.624,0	4.599,0	1.025,0	-	-	-	-
2007	5.728,0	4.703,0	1.025,0	-	-	-	-
2008	5.798,0	4.773,0	1.025,0	-	-	-	-
2009	6.389,7	5.401,0	987,0	1,7	-	-	-
2010	6.397,7	5.401,0	995,0	1,7	-	-	-
2011	6.350,0	5.352,0	998,0	-	-	-	-
2012	6.420,0	5.397,0	1.023,0	-	-	-	-
2013	7.353,3	6.230,1	1.082,5	2,7	1,0	37,0	-
2014	7.353,4	6.233,4	1.077,9	2,7	2,4	37,0	-
2015	7.806,7	6.652,8	1.103,4	7,7	4,8	37,0	1,0
2016	7.910,4	6.726,8	1.105,0	15,7	24,9	37,0	1,0
2017	7.941,5	6.748,0	1.106,4	15,7	28,4	42,0	1,0
2018	7.828,9	6.552,2	1.130,8	66,0	34,9	44,0	1,0
2019	7.641,6	6.350,3	1.144,8	66,1	35,4	44,0	1,0
2020	7.621,6	6.326,1	1.149,4	66,0	35,1	44,0	1,0
2021	7.965,2	6.649,4	1.157,2	66,0	47,9	44,0	0,7
2022	7.976,9	6.652,3	1.164,7	64,0	51,2	44,0	0,7
2023	7.907,4	6.323,1	1.209,4	64,0	266,2	44,0	0,7

The operating mode of the station is also characterized by the given power factor. This factor is measured by the ratio of the average annual load to the installed power of the station ($U_{m.power}$):

$$K_y = \frac{P_{or.oil}}{P_{m.güc}}$$

The operating mode of the station is also characterized by the given power factor. This factor is measured by the ratio of the average annual load to the installed power of the station ($U_{m.power}$).

$$K_{eh} = \frac{P_{m.güc}}{P_m}$$

At that time, taking into account the wind energy resources of Kalbajar and Lachin regions, the wind energy potential in the mountainous regions of Karabakh should be estimated at more than 500 megawatts, and the majority of this potential belongs to the border areas of the above-mentioned regions with Armenia. Despite such great potential, due to the lack of suitable infrastructure, it is necessary to carry out scientific-research works on the issue of obtaining electricity from wind energy in these areas. Sarsang HPP, built on the Tartar river in Tartar region, was commissioned in 1977 [11]. Sarsang hydroelectric complex facilities include an earthen dam, construction tunnel, emergency culvert, intake, diversion tunnel, HPP building and irrigation drawer unit. Two radial-axis (Francis system) RO115/697-B120 hydroturbines are installed at the station. Each of them has a power of 25 MW and a rotation speed of 375 rpm. The nominal head of these turbines is 87.5 m, and the water consumption is 33.25 m³/sec.

Taking into account that 25% of Azerbaijan's local water resources are formed in Karabakh, the prospects of using large rivers such as Tartar, Bazarchay, Hekari and their tributaries for the production of electricity are of great

importance for the liberated territories. Small hydropower plants operating in the region are mainly located in Tartar, Lachin and Kalbajar regions. Some small hydroelectric power stations have been repaired and restored.

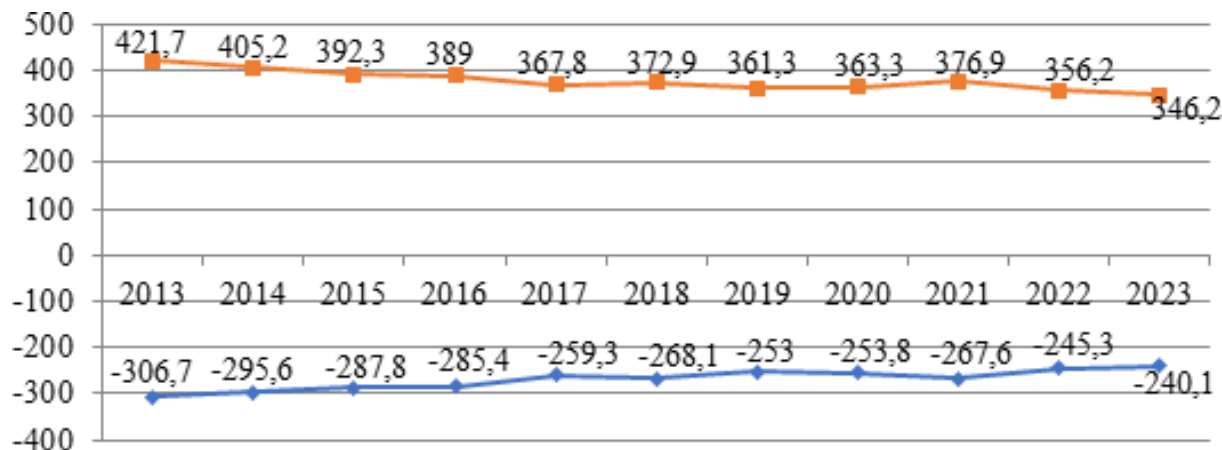


Fig. 2. Energy dependence and self-sufficiency in electricity at Sarsan hydroelectric power plant, in percentage

The 240 MW Shafaq Gunesh Power Station in Jabrayil district is one of the main steps taken towards the restoration and modernization of the electricity network in the liberated territories. The plant is the first industrial solar energy project in the liberated territories and the largest foreign direct investment project. The project was implemented on the basis of the Implementation Agreement signed between the Ministry of Energy and BP on June 3, 2021. Shafaq SEP is expected to produce about 500 million kilowatt-hours of electricity per year, which means saving 120-150 million cubic meters of natural gas and reducing one thousand tons of carbon emissions. Under the Virtual Power Transmission Agreement (VPTA) mechanism, the energy produced at the plant will be transferred to the existing electricity grid in Jabrayil and the same amount of energy will be supplied to the Sangachal terminal. As a result, renewable energy will be used to meet the terminal's energy needs and help reduce BP's operational emissions in the Caspian region [10].

Thus, the implementation of the project will make an important contribution to both the national economy and the decarbonization efforts of the consortium members, including SOCAR. Commissioning of another hydroelectric power station in the liberated territories is another positive step towards the restoration and reconstruction of electricity networks in these territories. "Azerenerji" OJSC's Sarikishlag HPP in Zangilan district is one of the 4 hydroelectric power plants built by cascade method on Okchu river in Zangilan massif. Before that, with the participation of the head of state, the remaining 3 hydropower stations - Jahangirbeyli, Shaiflin and Zangilan hydropower stations were commissioned.

All works in Sarikishlag HPP, built according to the Green Energy Zone concept, from the installation of guide pipes and foundation works to the construction of the machine room and the construction of the open distributor, were carried out in full compliance with the legal and regulatory acts. Special devices have been installed for the protection of flora and fauna, 35 percent of the water coming out of the main intake has been kept in the river channel, diverting pipes have been installed to ensure that the water coming out of the pipe flows back into the river without loss. Two eco-turbines that stop automatically when the water level in the river drops are installed in the engine room of Sarikishlag HPP, which is under construction. These turbines ensure that the river holds twice as much water as the ecological flow of the Okchu River. It was reported that a 110 kV open distribution facility was built at SES. This device performs the function of transferring "green energy" generated not only at Sarikishlag HPS, but also at Shayifli HPS to the system. Here, both the substation and the entire hydroelectric power station are equipped with digital equipment and an automatic control system. Sarikishlag will produce up to 33 million kilowatt-hours of electricity per year, thereby saving 7.4 million cubic meters of gas and preventing the release of 13 thousand tons of carbon dioxide into the atmosphere. The Zangilan waterfall with a capacity of 42 MW, consisting of 4 hydroelectric power stations, including the Sarikishlag HPP, is the largest cascade created in the territories freed from occupation.

Designed by local experts, the newly built 42 MW four cascade power plant operates at more than 70 percent of installed capacity. Part of the energy produced here is used for household needs, and the rest is sent to the country's energy grid. It also increases regional energy independence and contributes to environmental protection [11]. In general, unlike other renewable energy sources that depend on weather conditions, 32 hydropower plants with a capacity of 270 megawatts commissioned by "Azerenerji" in Karabakh and East Zangezur produce "green energy" day and night and without interruption. As a result of President Ilham Aliyev's resolute position on turning the region into a "green energy" zone, for the first time in the history of the country's energy system, Karabakh and East Zangezur regions have become regions with zero carbon emissions in all seasons, and the demand for electricity is fully covered by "green energy" thanks to the operating hydropower plants.

Emphasizing the significant solar energy potential of the southern plains of Karabakh, especially Fuzuli, Jabrayil and Zangilan regions, is one of the most important issues. It is emphasized here that these areas receive significant solar radiation, second only to Nakhchivan, with an annual solar radiation intensity of 1600-1700 kWh per square meter and a total energy potential of 3000-4000 megawatts. From an analytical point of view, this information suggests that

these regions can play a decisive role in Azerbaijan's renewable energy strategy. Noting measurement observation stations for accurate calculations after full land release shows commitment to data-driven development. It also means that previous conflicts may hinder the appreciation and use of this resource. Once the necessary infrastructure is created, the region can make a significant contribution to Azerbaijan's energy independence and sustainability efforts. Overall, the article highlights the strategic importance of these regions for solar energy production, both for national energy security and efficient use of renewable resources. This article also highlights the significant wind energy potential in the mountainous regions of Karabakh, especially Kalbajar and Lachin. It is noted that the average annual speed of the wind reaches 100 meters, it reaches 7-8 m/s in many regions, and it reaches 10 m/s in the border regions with Armenia. Estimated wind energy potential is 300-500 megawatts, with plans for additional measurements once the site is fully cleared. From an analytical point of view, these wind conditions offer strong opportunities for wind energy development, as wind speeds above 7 m/s are generally considered favorable for commercial wind farms. Mention of "Globalwindatlas.info" indicates reliance on an international database, but the text also acknowledges the need for on-site measurements to refine potential estimates. This reinforces the broader theme of post-conflict reconstruction, where renewable energy can play a key role in economic recovery and sustainable development. Investing in wind energy in these regions can expand Azerbaijan's energy mix, reduce dependence on fossil fuels and contribute to long-term energy security. However, successful implementation will require infrastructure development, network integration and investment incentives to attract stakeholders.

Overall, the article emphasizes the strategic importance of wind energy in Karabakh and the need for additional assessments to reveal its full potential. There are geothermal energy and coal resources in the Karabakh region, especially in Kalbajar, Shusha and Tartar. Geothermal potential shows thermal water reserves of 3093 m³/day in Kalbajar and 412 m³/day in Shusha. In addition, there are 8.5 million tons of coal reserves in Chardagli village of Tartar. From an analytical perspective, these resources present both opportunities and challenges. The presence of thermal waters indicates potential for the development of geothermal energy, which can be used for heat, electricity generation, or tourism-related applications such as the spa and wellness industry. However, the feasibility of large-scale geothermal energy production would require further exploration and technological investment. Chardali coal reserves highlight a traditional energy resource, but their use must be assessed in the context of environmental impact and Azerbaijan's broader energy transition goals. While coal can provide short-term energy growth, a long-term sustainable approach will favor cleaner energy sources such as solar, wind and geothermal.

The article provides general information on the energy potential of Jabrayil and Fuzuli regions, focusing on water and solar energy opportunities. Both regions are located in the southeastern part of the Lesser Caucasus, Jabrayil borders Iran along the Araz River, and Fuzuli stretches from the Karabakh Range to the river. Hydroelectric potential There is a compatibility of hydroelectric power stations (SES) along the Araz River in Fizuli and Jabrayl. This is a logical approach because rivers provide a sustainable source of energy and hydropower is a well-established renewable energy method. However, the article mentions "hydroelectric plants" in the context of solar energy, which seems like an inconsistency. It is likely to highlight solar power plants (SPPs) as a viable option. The paper proposes a combination of hydropower and solar energy as a balanced approach to energy diversification. Hydropower provides continuous baseload power, while solar power can supplement demand during peak hours of sunlight. The restoration of the electricity grid in Fuzuli highlights the need to rebuild the energy infrastructure after the conflict, with renewables playing a key role. Using upland and salt flats for hydro projects while allocating other land for solar energy is consistent with smart land use strategies. Jabrayil and Fuzuli regions have a strong potential in the development of renewable energy, especially in the field of water and solar energy. Detailed feasibility studies, investments in modern network infrastructure and careful environmental planning will be required to maximize efficiency. Proper implementation can significantly contribute to Azerbaijan's energy security and sustainability goals for these regions. In addition, as a result of the article, an overview of the energy potential of several regions of Karabakh is presented, emphasizing the development of hydro, geothermal and general infrastructure in the analysis. The geographical and climatic conditions of each region dictate the most suitable energy solutions.

Aghdam district, the district has a mixture of plains and mountains, but the main focus is on hydroelectricity. Considering that Karabakh is the most densely populated area, energy infrastructure is of crucial importance. The construction of "Aghdam-1" and "Aghdam-2" substations by "Azerenergy" indicates the work done in the direction of integration of the region into the general energy network of Azerbaijan and provision of stable and uninterrupted electricity supply. In the case of hydropower, Aghdam's partially plain relief and relatively low altitude compared to other regions may make large-scale hydropower projects less feasible. Solar energy can also be explored instead.

Kalbajar district, being the highest mountainous district, Kalbajar has high hydropower potential due to numerous water sources and steep elevation changes ideal for hydroelectric dams. Geothermal energy is worth mentioning. With existing thermal water resources (3,093 m³/day), there is potential for geothermal heating and electricity generation. However, further exploration is needed to determine the economic feasibility of geothermal power plants.

Lachin district, another mountainous region, has made hydroelectric power (KES - probably hydroelectric plants) a natural choice. Like Kalbajar, its elevation and river systems can support small to medium hydropower plants. However, infrastructure development can be challenging due to terrain and accessibility. Solar energy can be limited due to high altitude climatic conditions and snowfall. Khojaly region has similar characteristics to Kalbajar and Lachin, which has a mountainous terrain and significant altitude (up to 2843 m), which makes it the main option for hydropower. The presence of mountain-forest and mountain-meadow lands points to potential environmental problems that must be taken into account in the planning of energy projects.

Mention of the Badara and Gargar rivers indicate potential locations for hydro projects. Due to the mountainous terrain and water resources, the main source of renewable energy in these regions is hydropower. It is especially suitable for Kalbajar, Lachin and Khojaly water projects. Geothermal energy in Kalbajar is an interesting possibility, but further feasibility studies are needed. Aghdam stands out as an energy hub with its large population and planned substations and is strengthening its role in regional energy distribution. To maximize energy potential, environmental and technical challenges (flat terrain, cold climate, and infrastructure development) must be addressed. Each region has unique energy opportunities, with hydropower as the most favorable option, geothermal potential in Kalbajar, and grid infrastructure improvement in Agdam. A comprehensive energy strategy that takes into account geographical, climatic and infrastructural factors will be essential for sustainable development in Karabakh.

In addition, information is provided about the energy potential of Gubadli and Zangilan regions, hydroelectricity and solar energy opportunities are emphasized. Both regions have unique geographical features that make them suitable for renewable energy projects. Gubadli district, with an area of 802 km², Gubadli is surrounded by major districts such as Lachin, Zangilan and Jabrayil, as well as Armenia from the west. The region is rich in water resources, it has rivers such as Bazarchay (Barghusad), Hekari, Kichic Hekari, Meydan Daresi, and numerous springs. Given the availability of multiple water sources, hydroelectric power (SEP) is the most affordable energy option. Noting the construction of HPPs in areas unsuitable for agriculture shows an effort to optimize land use while minimizing agricultural disruption. Zangilan district is on the left bank of the Araz River, bordering Armenia (Sunik district) and Iran (East Azerbaijan province). Proximity to the Araz River, a major waterway, makes hydropower a logical choice. The region is notable for its high number of sunny days, which makes it suitable for solar power plants. Hydropower is the predominant renewable energy source in both Gubadli and Zangilan due to the presence of many rivers and the proximity to the Araz River. Zangila's solar energy potential cannot be overlooked as its climatic conditions make it an excellent candidate for solar farms. Land use planning is emphasized, ensuring that energy projects do not interfere with agriculture. The development of the infrastructure will be especially important for the integration of these renewable energy sources into the electricity grid of Azerbaijan.

Both Gubadli and Zangilan have strong hydropower potential, Zangilan is additionally suitable for solar energy. A mixed renewable energy strategy focusing on hydropower as a fixed base and solar as an additional source will maximize efficiency and sustainability in these regions. The energy potential of the cities of Shusha, Aghdara, Khankendi and Khojavand is described, due to their mountainous terrain and water resources, the main attention is paid to hydroelectric power. In addition, the potential of geothermal energy in Shusha is briefly mentioned. The city of Shusha, a mountainous area with steep cliffs, makes large-scale energy projects difficult. Hydropower is proposed, although feasibility depends on the presence of strong river currents. Geothermal energy is probably associated with underground heat sources or thermal waters. However, the text does not provide details on the existing geothermal activity, so further exploration should be done. Aghdara region, different topography - mountainous in most parts, plain in the east. Especially Tartar and Khachinchay are rich in water resources. The Sarsang Reservoir (built in 1976) already supports both electricity generation and irrigation, indicating the existing infrastructure for hydropower. Hydroelectric power plants (HEPs) are a logical choice given the established water resources.

The city of Khankendi and Gargachay have mountainous and foothill terrain. It suggests hydropower as the most suitable energy source, which is compatible with the availability of river-based energy. The location of the city in a strategic area means that the development of energy infrastructure here will be important for the stability and economic development of the region. Khojavand region is located in the Karabakh range, bordering Shusha with the Beyuk-Kirs peak (2725 m). A temperate-warm climate with seasonal dryness that can affect energy needs (eg, higher electricity demand for heating in winter). Many rivers and springs (tributaries of Kondalan, Ishgan, Gozlu, Guru and Hekari rivers) create a strong potential for hydropower. The connection to the Araz river basin creates the basis for the development of regional hydropower.

Due to abundant rivers and reservoirs, hydropower is the dominant energy source in all four regions. The existing infrastructure in Aghdara (Sarsang reservoir) creates a foundation for the future development of hydropower. Geothermal energy in Shusha is an interesting possibility, but more geological studies are needed to confirm its viability. Shusha and Khankendi are politically and economically important, requiring a reliable electrical infrastructure. Aghdara and Khojavand have strong water resources, which can provide not only local energy needs, but also the distribution of regional hydropower. Steep mountains in Shusha and Khojavand may complicate energy infrastructure projects. Seasonal drought periods can affect river flows, requiring hydro solutions based on reservoirs rather than river systems. Hydropower is the most affordable energy option for these regions using rivers and reservoirs. Shusha's geothermal potential needs to be further explored, but hydropower remains the main renewable source. The existing hydropower infrastructure (Sarsang reservoir) provides a strong basis for the expansion of electricity production. Strategic energy planning will be necessary to ensure stability, efficiency and sustainability in the post-conflict reconstruction of these regions.

The studied article deals with the issue of modeling and reconstruction of energy systems. Here, the priorities for the construction and development of energy systems in the regions freed from occupation have been determined. At the same time, the integration of hybrid energy systems into the reconstructed energy systems in these areas was noted as a priority issue. On the part of the country's leadership, it pays special attention to the reconstruction of electricity networks within the framework of large-scale restoration works in the territories freed from occupation. In particular, based on the "green energy" concept, priority is given to alternative energy sources in Karabakh and East Zangezur

regions. Building electricity networks in the liberated territories is a complex process from a strategic and technical point of view. For the successful implementation of this process, work should be done in several main directions.

Reconstruction of infrastructure;

- Application of "Smart Network" technologies;
- Development of alternative and green energy sources;
- Ensuring safety and stability of electricity supply;
- Energy supply for industry and agriculture;
- International partnership and investment.

The most favorable regions for the development of solar energy in the occupied territories are mainly regions with high solar radiation. Karabakh and Eastern Zangezur economic regions have great potential in this regard. Regions suitable for solar energy:

1. Cabrayil – It has vast plains and abundant solar radiation. There are ideal conditions for building solar power plants.
2. Zangilan - The potential of both solar and wind energy is high. There are large open spaces for installing solar panels.
3. Fuzuli – The number of sunny days is high and large areas are suitable for energy projects.
4. Aghdam is a region with potential for the use of renewable energy sources for industry and agriculture.
5. Gubadli – has a relief suitable for using solar energy.
6. Lachin – Although there is more potential for wind energy, there are also opportunities for solar energy.

These regions are considered a priority for the development of solar energy projects and are expected to attract more investments in the future. Existing and Planned Solar Stations:

- It is planned to build large solar power plants in Zangilan and Jabrayl.
- It is planned to create a large renewable energy zone and base Karabakh and Eastern Zangezur on "green energy" sources.

3. Conclusion

One of the most favorable regions for wind energy projects is Lachin region. The geographical position and climatic conditions of Lachin region provide high and continuous wind flow, which is important for the efficient operation of wind turbines. In addition, other regions of Eastern Zangezur economic region, for example, Zangilan and Gubadli regions, are also being investigated in terms of wind energy potential. Due to the stable wind speed and favorable relief conditions in these areas, it is planned to expand wind energy projects in these regions in the future. In general, the use of wind energy plays an important role in the development of the renewable energy sector in the region, and both local and international investments are being attracted to this area.

1. Reference

2. President of the Republic of Azerbaijan. (2013, February 11). *On additional measures in the field of alternative and renewable energy*.
3. President of the Republic of Azerbaijan. (2005, February 14). *The State Program for the Development of the Fuel-Energy Complex of the Republic of Azerbaijan (2005–2015)*.
4. President of the Republic of Azerbaijan. (1999, December 28). *The Law of the Republic of Azerbaijan on Electric and Thermal Power Plants*.
5. President of the Republic of Azerbaijan. (1995, April 3). *The Law of the Republic of Azerbaijan on Electric Energy*.
6. Hajizadeh, E. (2008). National energy strategy: Conceptual justifications and perspectives. *Revival-XXI Century*, (124–125), 283.
7. Hajizadeh, E. M. (2000). *The Energy Complex is on the Verge of New Reforms*. Baku.
8. Guliyev, H. B. (2022). *Development of the Theoretical and Methodological Bases of the Management of the Quality Indicators of Electric Energy in Electric Power Systems Based on Intellectual Technologies*. Baku.
9. Azenerji. (n.d.). *About the Development of Energy in the Republic of Azerbaijan*. Retrieved from <http://azenerji.com/az/information/articles/article1.html>
10. State Statistical Committee of the Republic of Azerbaijan. (n.d.). Retrieved from <https://stat.gov.az/>
11. Azerenerji. (n.d.). Retrieved from <https://azerenerji.gov.az/>

12. Abaszade, R. G., Kapush, O. A., Nabiyev, A. M. (2020). Properties of carbon nanotubes doped with gadolinium. *Journal of Optoelectronic and Biomedical Materials*, 12(3), 61–65. Retrieved from https://www.chalcogen.ro/61_AbaszadeRG.pdf
13. Abaszade, R. G., Kapush, O. A., Mamedova, S. A., Nabiyev, A. M., Melikova, S. Z., Budzulyak, S. I. (2020). Gadolinium doping influence on the properties of carbon nanotubes. *Physics and Chemistry of Solid State*, 21(3), 404-408. <https://doi.org/10.15330/pcss.21.3.404-408>
14. Figarova, S. R., Aliyev, E. M., Abaszade, R. G., Alekberov, R. I., Figarov, V. R. (2021). Negative differential resistance of graphene oxide/sulfur compound. *Journal of Nano Research*, 67, 25-31. <https://doi.org/10.4028/www.scientific.net/JNanoR.67.25>
15. Abaszade, R. G., Mammadova, S. A., Kapush, O. A., Agayev, F. G., Nabiyev, A. M., Mammadova, M. M., Budzulyak, S. I., Kotsyubynsky, V. O. (2021). Synthesis and characterization of graphene oxide flakes for transparent thin films. *Physics and Chemistry of Solid State*, 23(3), 595-601. <https://doi.org/10.15330/pcss.22.3.595-601>
16. Mammadov, A. G., Abaszade, R. G., Kotsyubynsky, V. O., Gur, E. Y., Bayramov, I. Y., Khanmammadova, E. A., Kapush, O. A. (2022). Photoconductivity of carbon nanotubes. *Technical and Physical Problems of Engineering*, 14(3), 155-160.
17. Abaszade, R. G. (2022). Synthesis and analysis of flakes graphene oxide. *Journal of Optoelectronic and Biomedical Materials*, 14(3), 107–114. <https://doi.org/10.15251/JOBM.2022.143.107>
18. Abaszade, R. G., Mammadov, A. G., Bayramov, I. Y., Khanmammadova, E. A., Kotsyubynsky, V. O., Gur, E. Y. (2022). Structural and electrical properties of sulfur-doped graphene oxide/graphite oxide composite. *Physics and Chemistry of Solid State*, 25(2), 256-260. <https://doi.org/10.15330/pcss.23.2.256-260>
19. Abaszade, R. G., Mammadov, A. G., Bayramov, I. Y., Khanmammadova, E. A., Kotsyubynsky, V. O., Gur, E. Y., Kapush, O. A., Boychuk, V. M. (2022). Modeling of voltage-ampere characteristic structures on the basis of graphene oxide/sulfur compounds. *Technical and Physical Problems of Engineering*, 14(2), 302-306.
20. Boychuk, V. M., Zapukhlyak, R. I., Abaszade, R. G., Kotsyubynsky, V. O., Hodlevsky, M. A., Rachiy, B. I., Turovska, L. V., Dmytriv, A. M., Fdorchenko, S. V. (2022). Solution combustion synthesized NiFe₂O₄/reduced graphene oxide composite nanomaterials: Morphology and electrical conductivity. *Physics and Chemistry of Solid State*, 23(2), 815–824. <https://doi.org/10.15330/pcss.23.4.815-824>
21. Abaszade, R. G., Aliyev, E. M., Babanli, M. B., Kotsyubynsky, V. O., Zapukhlyak, R. I., Mammadov, A. G., Budak, H. F., Kasapoglu, A. E., Gur, E., Margitych, T. O., Stetsenko, M. O. (2023). Investigation of thermal properties of carbon nanotubes and carboxyl group-functionalized carbon nanotubes. *Physics and Chemistry of Solid State*, 24(3), 530–535. <https://doi.org/10.15330/pcss.24.3.530-535>
22. Guliyeva, N. A., Abaszade, R. G., Khanmammadova, E. A., Azizov, E. M. (2023). Synthesis and analysis of nanostructured graphene oxide. *Journal of Optoelectronic and Biomedical Materials*, 15(1), 23–30. <https://doi.org/10.15251/JOBM.2023.151.23>
23. Figarova, S. R., Aliyev, E. M., Abaszade, R. G., Figarov, V. R. (2023). Negative thermal expansion of sulfur-doped graphene oxide. *Advanced Materials Research*, 1175, 55–62. <https://doi.org/10.4028/p-rppn12>
24. Abaszade, R. G., Babanli, M. B., Kotsyubynsky, V. O., Mammadov, A. G., Gur, E., Kapush, O. A., Stetsenko, M. O., Zapukhlyak, R. I. (2023). Influence of gadolinium doping on structural properties of carbon nanotubes. *Physics and Chemistry of Solid State*, 24(1), 153–158. <https://doi.org/10.15330/pcss.24.1.153-158>
25. Mammadov, A. G., Abaszade, R. G., Babanli, M. B., Kotsyubynsky, V. O., Gur, E., Soltabayev, B. D., Margitych, T. O., Stetsenko, M. O. (2023). Photoconductivity of gadolinium-doped carbon nanotubes. *International Journal on Technical and Physical Problems of Engineering*, 15(3), 53–58.
26. Abaszade, R. G., Mammadova, A. G., Khanmammadova, E. A., Bayramov, I. Y., Namazov, R. A., Popal, Kh. M., Melikova, S. Z., Qasimov, R. C., Bayramov, M. A., Babayeva, N. İ. (2023). Electron paramagnetic resonance study of gadolinium-doped graphene oxide. *Journal of Ovonic Research*, 19(2), 259–263. <https://doi.org/10.15251/JOR.2023.193.259>
27. Stetsenko, M. O., Abaszade, R. G. (2023). X-ray phase analysis of carbon nanotubes obtained by the arc discharge method. *UNEC Journal of Engineering and Applied Sciences*, 3(1), 15–20. <https://doi.org/10.61640/ujeas.2023.0503>
28. Abaszade, R. G., Aliyev, E. M., Mammadov, A. G., Khanmammadova, E. A., Guliyev, A. A., Aliyev, F. G., Zapukhlyak, R. I., Budak, H. F., Kasapoglu, A. E., Margitych, T. O., Singh, A., Arya, S., Gur, E., Stetsenko, M. O. (2024). Investigation of thermal properties of gadolinium-doped carbon nanotubes. *Physics and Chemistry of Solid State*, 25(1), 142–147. <https://doi.org/10.15330/pcss.25.1.142-147>
29. Sharma, B., Singh, A., Sharma, A., Dubey, A., Gupta, V., Abaszade, R. G., Sundramoorthy, A. K., Sharma, N., Arya, S. (2024). Synthesis and characterization of zinc selenide/graphene oxide (ZnSe/GO) nanocomposites for electrochemical detection of cadmium ions. *Applied Physics A*, 130(297). <https://doi.org/10.1007/s00339-024-07472-0>
30. Abaszade, R. G., Mammadov, A. G., Khanmammadova, E. A., Aliyev, F. G., Kotsyubynsky, V. O., Gur, E., Soltabayev, B. D., Margitych, T. O., Stetsenko, M. O., Singh, A., Arya, S. (2024). Photoconductivity of functionalized carbon nanotubes. *Digest Journal of Nanomaterials and Biostructures*, 19(2), 837–843. <https://doi.org/10.15251/DJNB.2024.192.837>

31. Ivanichok, N., Kolkovskiy, P., Ivanichok, O., Kotsyubynsky, V., Boychuk, V., Rachiy, B., Bembenek, M., Warguła, Ł., Abaszade, R., Ropyak, L. (2024). Effect of thermal activation on the structure and electrochemical properties of carbon material obtained from walnut shells. *Materials*, 17(2514), 1–21. <https://doi.org/10.3390/ma17112514>
32. Zapukhlyak, R. I., Kotsyubynsky, V. O., Boychuk, V. M., Rachiy, B. I., Abaszade, R. G., Hoi, V. T., Klymyuk, M. (2025). Structural evolution of porous carbon materials derived from hemp fibers: Raman spectroscopy studies. *Physics and Chemistry of Solid State*, 26(1), 132–139. <https://doi.org/10.15330/pcss.26.1.132-139>
33. Moradi, R., Pour Khalili, N., Khanmamadova, E., Abaszade, R. (2024). Functionalized carbon nanostructures for flexible electronics. In A. Barhoum & K. Deshmukh (Eds.), *Handbook of functionalized carbon nanostructures* (pp. 2581–2614). Springer. https://doi.org/10.1007/978-3-031-14955-9_71-1
34. Khanmamadova, E., Abaszade, R., Moradi, R. (2025). Hydrophobic and hydrophilic behavior of multifunctional thin films. In *Multi-scale and multifunctional coatings and interfaces for tribological contacts* (pp. 83–102). <https://doi.org/10.1201/9781032635347-6>
35. Khanmamadova, E., Abaszade, R., Moradi, R. (2025). Antiviral thin surface and surface life. In *Multi-scale and multifunctional coatings and interfaces for tribological contacts* (pp. 121–134). <https://doi.org/10.1201/9781032635347-8>
36. Abaszade, R., Khanmamadova, E. (2025). Specific applications for unmanned aerial vehicles (UAVs) and aerial robotics. In S. Khalid & S. Verma (Eds.), *Advanced antenna technologies for aerial platforms: From design to deployment* (p. 435). <https://doi.org/10.4018/979-8-3693-6035-4>
37. Khanmamadova, E., Abaszade, R. (2025). Advanced design, materials, and manufacturing techniques for next-generation antennas. In S. Khalid & S. Verma (Eds.), *Advanced antenna technologies for aerial platforms: From design to deployment* (p. 435). <https://doi.org/10.4018/979-8-3693-6035-4>
38. Khanmamadova, E., Abaszade, R. (2025). The evolution of green mobility: A critical comparison of pure electric and hybrid electric vehicles. In S. Khalid, J. Ali, & D. S. Yadav (Eds.), *Hybrid electric vehicles and distributed renewable energy conversion: Control and vibration analysis* (pp. 251–187). <https://doi.org/10.4018/979-8-3693-5797-2.ch011>

A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a cloudy sky. The image is framed by a dark green curved border at the top.

eco

ENERGETIKA
ENERGETICS

A graphic of a water splash or wave, rendered in a soft, painterly style, located in the bottom right corner of the page.